Bézier Curve — Algebraic Expansion

Explanation of the Program Output

Case 2: Quartic 3-D, 5 Control Points

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This document explains the output of `01-bezier-algebra.jl` when run with `CASE = 2`. It walks through every printed block, explains what each line means, and verifies the key numbers by hand. No programming knowledge is required — only basic algebra.

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1. Background: What Is a Bézier Curve?

A **Bézier curve** is a smooth curve controlled by a set of *control points*. You choose the points; the curve is pulled toward them like magnets, but does not necessarily pass through them (except the first and last).

The curve is parameterised by a single number $u \in [0, 1]$. At $u = 0$ you are at the start of the curve; at $u = 1$ you are at the end. Every value in between gives you a point somewhere along the curve.

The mathematical formula is:

$$\mathbf{P}(u) = \sum_{i=0}^{n} B(n, i, u)\ P_i$$

where $P_0, \ldots, P_n$ are the control points and $B(n, i, u) = \binom{n}{i} u^i (1-u)^{n-i}$ is the i-th **Bernstein basis polynomial**.

The program's job is to expand this formula into ordinary polynomials in u (one per axis), then evaluate them.

2. Header and Control Points

```
================================================================
  Bezier Curve - Algebraic Expansion
  CASE 2  |  Quartic  |  3D  |  5 Control Points
================================================================

  Control Points:
    P0 :  x=0  y=0  z=1
    P1 :  x=0  y=4  z=1
    P2 :  x=4  y=0  z=1
    P3 :  x=4  y=4  z=1
    P4 :  x=5  y=4  z=1
```

Quartic means the curve polynomial has degree 4. Degree = (number of control points) $-1 = 5 - 1 = 4$.

The five control points in the xy-plane (ignoring z for a moment) are roughly:

Point	x	y
P_0	0	0
P_1	0	4
P_2	4	0
P_3	4	4
P_4	5	4

Key observation about z: Every single control point has $z = 1$. This means the entire curve lies in the flat plane $z = 1$. No matter what u is, $z(u)$ will always equal 1. We will see this confirmed in Step 4.

3. Step 1 — Bernstein Form

```
P(u) =  [x(u)  y(u)  z(u)]

P(u) =  B(4,0)*P0 + B(4,1)*P1 + B(4,2)*P2
                  + B(4,3)*P3 + B(4,4)*P4

where:
  B(4,0,u)  =  (1-u)^4
  B(4,1,u)  =  4u(1-u)^3
  B(4,2,u)  =  6u^2(1-u)^2
  B(4,3,u)  =  4u^3(1-u)
  B(4,4,u)  =  u^4
```

This is the curve formula before any algebra has been done. The curve point at parameter u is a *weighted average* of all five control points, where the weights are the five Bernstein polynomials.

Why these particular weights? Each $B(4, i, u)$ is large near $u = i/4$ and small elsewhere. So $B(4, 0)$ dominates near $u = 0$ (pulling the curve toward P_0), $B(4, 4)$ dominates near $u = 1$ (pulling toward P_4), and so on.

Boundary behaviour — easily checked without any algebra:

At $u = 0$: $(1 - 0)^4 = 1$, all other terms contain u so they are 0.

$$\mathbf{P}(0) = 1 \cdot P_0 = (0, 0, 1)$$

The curve **starts** exactly at P_0.

At $u = 1$: $1^4 = 1$, all other terms contain $(1 - u)$ so they are 0.

$$\mathbf{P}(1) = 1 \cdot P_4 = (5, 4, 1)$$

The curve **ends** exactly at P_4.

The intermediate points P_1, P_2, P_3 are *not* on the curve — they only attract it.

Partition of unity. The five weights always sum to exactly 1 for any u:

$$B(4, 0) + B(4, 1) + B(4, 2) + B(4, 3) + B(4, 4) = 1$$

This guarantees the curve stays inside the bounding box of the control points and behaves like a proper weighted average.

4. Step 2 — Substituting the Coordinates

```
x(u) =  (1-u)^4*0 + 4u(1-u)^3*0 + 6u^2(1-u)^2*4
                  + 4u^3(1-u)*4  + u^4*5
y(u) =  (1-u)^4*0 + 4u(1-u)^3*4 + 6u^2(1-u)^2*0
                  + 4u^3(1-u)*4  + u^4*4
z(u) =  (1-u)^4*1 + 4u(1-u)^3*1 + 6u^2(1-u)^2*1
                  + 4u^3(1-u)*1  + u^4*1
```

The program reads the x, y, z columns of the control point matrix and substitutes each coordinate as the scalar weight for its basis function.

Reading the x column: coordinates are $0, 0, 4, 4, 5$.

The first two basis terms are multiplied by 0, so they vanish. Only the last three matter:

$$x(u) = 6u^2(1-u)^2 \cdot 4 + 4u^3(1-u) \cdot 4 + u^4 \cdot 5$$

Reading the y column: coordinates are $0, 4, 0, 4, 4$.

The first and third terms vanish:

$$y(u) = 4u(1-u)^3 \cdot 4 + 4u^3(1-u) \cdot 4 + u^4 \cdot 4$$

Reading the z column: all coordinates are 1.

Every basis term gets weight 1, so:

$$z(u) = B(4,0) \cdot 1 + B(4,1) \cdot 1 + B(4,2) \cdot 1 + B(4,3) \cdot 1 + B(4,4) \cdot 1 = 1$$

(The five Bernstein polynomials sum to 1, so $z(u) = 1$ identically.)

5. Step 3 — Expanding Each Basis Term

```
B(4,0,u)  =  (1-u)^4   =  1 - 4u + 6u^2 - 4u^3 + u^4
B(4,1,u)  =  4u(1-u)^3 =  4u - 12u^2 + 12u^3 - 4u^4
B(4,2,u)  =  6u^2(1-u)^2  =  6u^2 - 12u^3 + 6u^4
B(4,3,u)  =  4u^3(1-u)  =  4u^3 - 4u^4
B(4,4,u)  =  u^4
```

Each Bernstein polynomial (which contains a product like $u^i(1-u)^{n-i}$) is multiplied out into a plain polynomial. These are the monomial expansions — each is now just a sum of powers of u.

Verify $B(4,0) = (1-u)^4$ by hand:

Using the binomial theorem $(a+b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$ with $a = 1$

and $b = -u$:

$$(1-u)^4 = 1 - 4u + 6u^2 - 4u^3 + u^4 \checkmark$$

Verify $B(4,2) = 6u^2(1-u)^2$ by hand:

$$(1-u)^2 = 1 - 2u + u^2$$

$$6u^2(1 - 2u + u^2) = 6u^2 - 12u^3 + 6u^4 \checkmark$$

Verify $B(4,1) = 4u(1-u)^3$ by hand:

$$(1-u)^3 = 1 - 3u + 3u^2 - u^3$$

$$4u(1 - 3u + 3u^2 - u^3) = 4u - 12u^2 + 12u^3 - 4u^4 \checkmark$$

The coefficients of $(1-u)^n$ follow Pascal's triangle with alternating signs: $1, -n, \binom{n}{2}, -\binom{n}{3}, \ldots$

For $n = 4$: $1, -4, +6, -4, +1$.

6. Step 4 — Collecting Terms

```
x(u):
  B(4,0)*   0  =  0
  B(4,1)*   0  =  0
  B(4,2)*   4  =  24u^2 - 48u^3 + 24u^4
  B(4,3)*   4  =  16u^3 - 16u^4
  B(4,4)*   5  =  5u^4
Collected:  x(u)  =  24u^2 - 32u^3 + 13u^4

y(u):
  B(4,0)*   0  =  0
  B(4,1)*   4  =  16u - 48u^2 + 48u^3 - 16u^4
  B(4,2)*   0  =  0
  B(4,3)*   4  =  16u^3 - 16u^4
  B(4,4)*   4  =  4u^4
Collected:  y(u)  =  16u - 48u^2 + 64u^3 - 28u^4

z(u):
  (all contributions)
Collected:  z(u)  =  1
```

This is the core algebraic result. Each expanded basis polynomial is scaled by its coordinate value, then all five are added together, collecting coefficients by power of u.

6.1. Collecting $x(u)$

The three non-zero contributions to $x(u)$, written as columns by power of u:

Contribution	u^2	u^3	u^4
$B(4,2) \times 4 = (6u^2 - 12u^3 + 6u^4) \times 4$	$+24$	-48	$+24$
$B(4,3) \times 4 = (4u^3 - 4u^4) \times 4$	0	$+16$	-16
$B(4,4) \times 5 = u^4 \times 5$	0	0	$+5$
Sum	**24**	**-32**	**13**

$$x(u) = 24u^2 - 32u^3 + 13u^4$$

Note: there is no constant term and no u^1 term because P_0 and P_1 both have $x = 0$, so they contribute nothing to $x(u)$.

6.2. Collecting $y(u)$

Contribution	u^1	u^2	u^3	u^4
$B(4,1) \times 4 = (4u - 12u^2 + 12u^3 - 4u^4) \times 4$	$+16$	-48	$+48$	-16
$B(4,3) \times 4 = (4u^3 - 4u^4) \times 4$	0	0	$+16$	-16
$B(4,4) \times 4 = u^4 \times 4$	0	0	0	$+4$
Sum	**16**	**-48**	**64**	**-28**

$$y(u) = 16u - 48u^2 + 64u^3 - 28u^4$$

6.3. Collecting $z(u)$

All five z-coordinates are 1, so all five basis polynomials are added with weight 1. Because they sum to 1 (partition of unity):

$$z(u) = 1$$

This is a degree-0 polynomial — a constant. No matter where you are on the curve, z is always 1.

6.4. Summary of Final Polynomials

Axis	Final polynomial
$x(u)$	$24u^2 - 32u^3 + 13u^4$
$y(u)$	$16u - 48u^2 + 64u^3 - 28u^4$
$z(u)$	1

These three polynomials fully describe the curve. To find any point on the curve, just substitute a value of u.

7. Step 5 — Evaluation Table

```
u        x(u)        y(u)        z(u)
```

```
--------------------------------------------
  0.00    0.000000    0.000000    1.000000
  0.25    1.050781    1.890625    1.000000
  0.50    2.812500    2.250000    1.000000
  0.75    4.113281    3.140625    1.000000
  1.00    5.000000    4.000000    1.000000
```

Each row is one sampled point on the curve. The program evaluates the three polynomials at $u = 0, 0.25, 0.5, 0.75, 1$.

7.1. Verifying the Boundary Points

At $u = 0$:

Substituting $u = 0$ into $24u^2 - 32u^3 + 13u^4$: every term contains u, so $x(0) = 0$. Similarly $y(0) = 0$.

$$\mathbf{P}(0) = (0,\ 0,\ 1) = P_0 \checkmark$$

The curve starts exactly at the first control point.

At $u = 1$:

$$x(1) = 24(1) - 32(1) + 13(1) = 24 - 32 + 13 = 5$$

$$y(1) = 16(1) - 48(1) + 64(1) - 28(1) = 16 - 48 + 64 - 28 = 4$$

$$\mathbf{P}(1) = (5,\ 4,\ 1) = P_4 \checkmark$$

The curve ends exactly at the last control point.

7.2. Verifying $u = 0.25$

Powers needed: $0.25^2 = 0.0625,\ \ 0.25^3 = 0.015625,\ \ 0.25^4 = 0.00390625.$

$x(0.25)$:

$$\begin{aligned} x(0.25) &= 24(0.0625) - 32(0.015625) + 13(0.00390625) \\ &= 1.5 - 0.5 + 0.05078125 \\ &= 1.05078125 \approx 1.050781 \checkmark \end{aligned}$$

$y(0.25)$:

$$\begin{aligned} y(0.25) &= 16(0.25) - 48(0.0625) + 64(0.015625) - 28(0.00390625) \\ &= 4 - 3 + 1 - 0.109375 \\ &= 1.890625 \checkmark \end{aligned}$$

7.3. Verifying $u = 0.50$

Powers needed: $0.5^2 = 0.25,\ \ 0.5^3 = 0.125,\ \ 0.5^4 = 0.0625.$

$x(0.50)$:

$$\begin{aligned} x(0.5) &= 24(0.25) - 32(0.125) + 13(0.0625) \\ &= 6 - 4 + 0.8125 \\ &= 2.8125\checkmark \end{aligned}$$

$y(0.50)$:

$$\begin{aligned} y(0.5) &= 16(0.5) - 48(0.25) + 64(0.125) - 28(0.0625) \\ &= 8 - 12 + 8 - 1.75 \\ &= 2.25\checkmark \end{aligned}$$

7.4. Verifying $u = 0.75$

Powers needed: $0.75^2 = 0.5625,\ \ 0.75^3 = 0.421875,\ \ 0.75^4 = 0.31640625.$

$x(0.75)$:

$$\begin{aligned} x(0.75) &= 24(0.5625) - 32(0.421875) + 13(0.31640625) \\ &= 13.5 - 13.5 + 4.11328125 \\ &= 4.11328125 \approx 4.113281\checkmark \end{aligned}$$

$y(0.75)$:

$$\begin{aligned} y(0.75) &= 16(0.75) - 48(0.5625) + 64(0.421875) - 28(0.31640625) \\ &= 12 - 27 + 27 - 8.859375 \\ &= 3.140625\checkmark \end{aligned}$$

8. Verification Block

```
Verification at u = 0.50:
  x(0.50)  =  24u^2 - 32u^3 + 13u^4
           =  24*0.50^2 + -32*0.50^3 + 13*0.50^4
           =  2.812500

  y(0.50)  =  16u - 48u^2 + 64u^3 - 28u^4
           =  16*0.50^1 + -48*0.50^2 + 64*0.50^3 + -28*0.50^4
           =  2.250000

  z(0.50)  =  1
           =  1
           =  1.000000
```

The program automatically picks the middle u value (0.50) and prints the full term-by-term substitution for each axis. This lets you follow the arithmetic step by step.

The z axis shows the simplest possible verification: $z(u) = 1$ is just a constant, so there is nothing to substitute — it is always 1 regardless of u.

9. Geometric Picture

Putting the five sampled points together:

u	x	y	z
0.00	0.000	0.000	1
0.25	1.051	1.891	1
0.50	2.813	2.250	1
0.75	4.113	3.141	1
1.00	5.000	4.000	1

What does the curve look like?

All five points have $z = 1$, so the curve is entirely flat — it lives in a single horizontal plane. In the xy-plane the curve starts at the origin $(0, 0)$, initially heads upward and to the right, reaches a midpoint near $(2.8, 2.3)$, then curves further right and upward toward $(5, 4)$.

The curve does *not* pass through $P_1 = (0, 4)$, $P_2 = (4, 0)$, or $P_3 = (4, 4)$. Those points pull the curve but the curve only interpolates (passes through) its two endpoints P_0 and P_4.

Between $u = 0.25$ and $u = 0.75$ the curve covers the middle portion of the journey. Notice that y grows more slowly near $u = 0.5$ (from 1.891 to 2.250, a rise of 0.359) compared to the first quarter (0 to 1.891). This reflects how the control point geometry shapes the speed at which the curve travels through space.